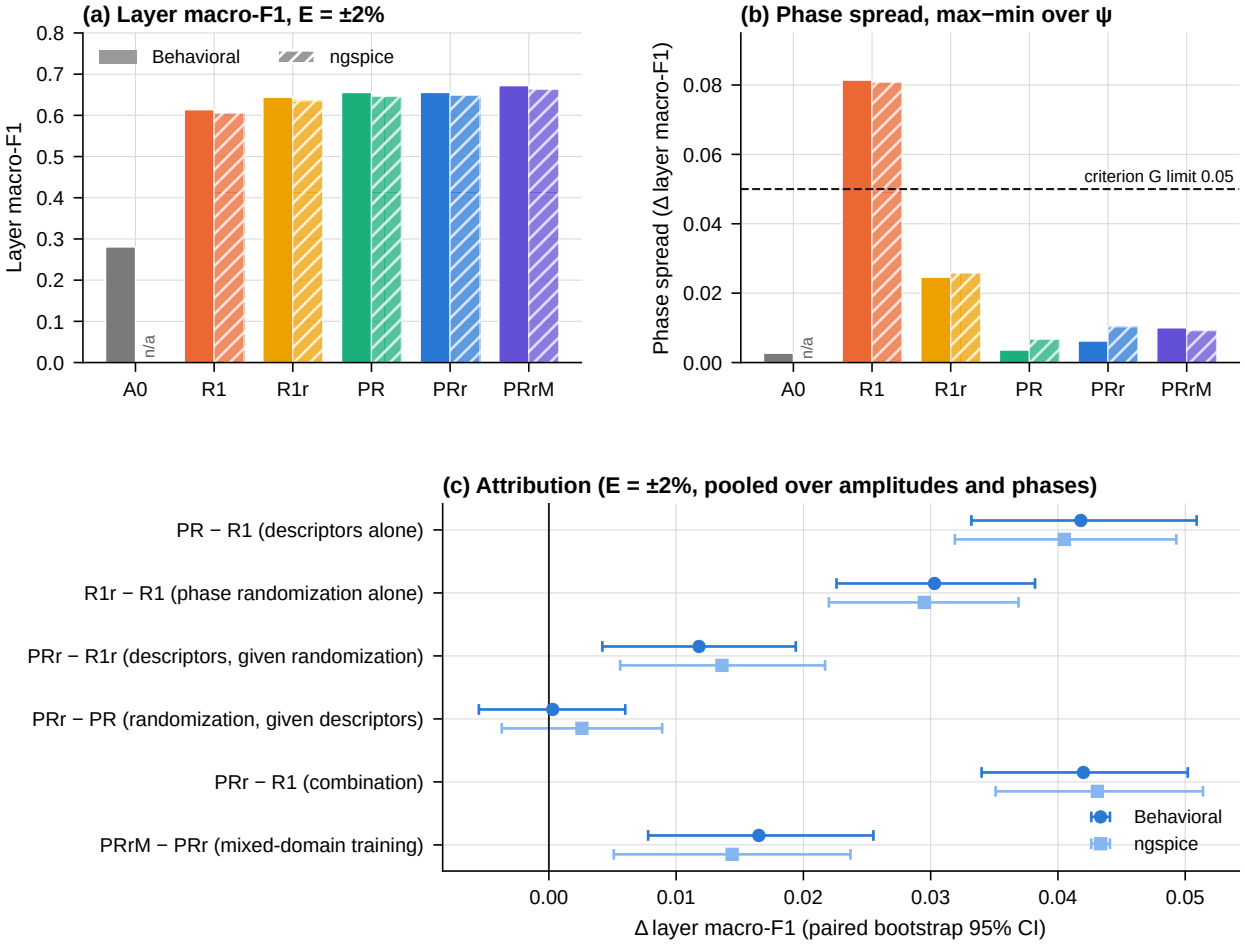
(a) Layer macro-F1, E = ±2%
Behavioral
ngspice
Layer macro-F1
n/a
A0
R1
R1r
PR
PRr
PRrM
(b) Phase spread, max−min over ψ
Phase spread (Δ layer macro-F1)
criterion G limit 0.05
(c) Attribution (E = ±2%, pooled over amplitudes and phases)
PR − R1 (descriptors alone)
R1r − R1 (phase randomization alone)
PRr − R1r (descriptors, given randomization)
PRr − PR (randomization, given descriptors)
PRr − R1 (combination)
PRrM − PRr (mixed-domain training)
Δ layer macro-F1 (paired bootstrap 95% CI)